随机模拟方法第四次作业

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1. 设 $X \sim \mathcal{U}[0,1]$, 并且 f 单调. 求证: $\mathrm{Cov}(f(X), f(1-X)) \leqslant 0$.

解答: 由 $f(x)$ 的单调性可得

$$\begin{aligned}
\mathrm{Cov}(f(X), f(1-X)) &= \mathbb{E}(f(X)f(1-X)) - \mathbb{E}f(X)\mathbb{E}f(1-X) \\
&= \int_0^1 f(x)f(1-x)\,\mathrm{d}x - \int_0^1 f(x)\,\mathrm{d}x \int_0^1 f(1-x)\,\mathrm{d}x \\
&= \frac{1}{2}\int_0^1 \int_0^1 [f(x)f(1-x) + f(y)f(1-y) - f(x)f(1-y) - f(y)f(1-x)]\,\mathrm{d}x\,\mathrm{d}y \\
&= \frac{1}{2}\int_0^1 \int_0^1 (f(x)-f(y))(f(1-x)-f(1-y))\,\mathrm{d}x\,\mathrm{d}y \leqslant 0.
\end{aligned}$$

□

2. 求证:

$$\mathrm{Var}(f(X)) = \mathrm{Var}(\mathbb{E}(f(X) \mid X^{(2)})) + \mathbb{E}(\mathrm{Var}(f(X) \mid X^{(2)})).$$

解答: 下面证明更一般的命题: 设 X, Y 是随机变量 (向量), 满足 $\mathbb{E}(f(X)^2) < +\infty$, 则

$$\mathrm{Var}(f(X)) = \mathrm{Var}(\mathbb{E}(f(X) \mid Y)) + \mathbb{E}(\mathrm{Var}(f(X) \mid Y)).$$

事实上,

$$\begin{aligned}
&\mathrm{Var}(\mathbb{E}(f(X) \mid Y)) + \mathbb{E}(\mathrm{Var}(f(X) \mid Y)) \\
&= \mathbb{E}(\mathbb{E}(f(X) \mid Y)^2) - \mathbb{E}(\mathbb{E}(f(X) \mid Y))^2 + \mathbb{E}(\mathbb{E}(f(X)^2 \mid Y) - \mathbb{E}(f(X) \mid Y)^2) \\
&= \mathbb{E}(\mathbb{E}(f(X) \mid Y)^2) - \mathbb{E}(f(X))^2 + \mathbb{E}(f(X)^2) - \mathbb{E}(\mathbb{E}(f(X) \mid Y)^2) \\
&= \mathbb{E}(f(X)^2) - \mathbb{E}(f(X))^2 \\
&= \mathrm{Var}(f(X)).
\end{aligned}$$

□

3. 设 f, g 是概率密度函数且 $g(x) > 0$. 求证: 相对熵 $D(f||g) \geqslant 0$, 并且等号成立当且仅当 $f(x) = g(x)$.

解答: 令 $\varphi(t) := t \ln t, t > 0$, 以及 $\varphi(0) = 0$. 由 Jensen 不等式,

$$D(f||g) = \int_{\mathbb{R}} f(x) \ln \frac{f(x)}{g(x)} \, \mathrm{d}x = \int_{\mathbb{R}} g(x) \varphi\left(\frac{f(x)}{g(x)}\right) \mathrm{d}x \geqslant \varphi\left(\int_{\mathbb{R}} g(x) \cdot \frac{f(x)}{g(x)} \, \mathrm{d}x\right) = \varphi\left(\int_{\mathbb{R}} f(x) \, \mathrm{d}x\right) = 0,$$

等号成立当且仅当 $\dfrac{f(x)}{g(x)} = C$, a.e., 再结合 $\int_{\mathbb{R}} f(x) \, \mathrm{d}x = \int_{\mathbb{R}} g(x) \, \mathrm{d}x = 1$ 可得等价于 $f(x) = g(x)$, a.e.. □